

Angle between lines

Key Points

To find the angle between two lines r_1 and r_2 find the angle between the two directions of the lines

d_1 and d_2 by

$$\cos \theta = \frac{d_1 \cdot d_2}{|d_1||d_2|}$$

Basic Skills

Q1. State the direction vector of the following lines

a) $r = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \end{pmatrix}$

b) $r = \begin{pmatrix} 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 5 \end{pmatrix}$

c) $r = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$

d) $r = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} - \mu \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

e) $r = \begin{pmatrix} 1 + \lambda \\ \lambda \\ 3\lambda \end{pmatrix}$

f) $r = \begin{pmatrix} 2t \\ 3 \\ 2 - t \end{pmatrix}$

Q2. Find the angle between the vectors

a) $a = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $b = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$

b) $a = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$ and $b = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

Competency Skills

Q3. Given the two lines $r_1 = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -1 \\ 1 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 7 \\ 7 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

- a)** State the two directions
- b)** Find the acute angle between the two directions
- c)** Hence find the obtuse angle between the lines

Q4. Find the acute angle between the following pairs of lines

a) $r_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$

b) $r_1 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 10 \\ -2 \end{pmatrix}$ and $r_2 = \begin{pmatrix} -1 \\ -3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ -10 \\ 2 \end{pmatrix}$

c) $r_1 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ -4 \end{pmatrix}$ and $r_2 = \begin{pmatrix} -1 \\ -3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

d) $l_1: \frac{x-6}{2} = \frac{y-3}{-2} = \frac{z}{4}$ and $l_2: \frac{x-1}{2} = \frac{2y}{-3} = \frac{z-5}{2}$

Advanced Skills

Q5. Given that the angle between $r_1 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} k \\ 1 \\ 1 \end{pmatrix}$ is 60° find the values of k

Q6. Find k such that the angle is minimal between $r_1 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3k \\ 2 \\ 1 \end{pmatrix}$ and $r_2 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$

Q7. A spiral in 2D is given by $r = e^t \begin{pmatrix} 1 + \sin t \\ \cos t \end{pmatrix}$

- a)** Find the points at which the spiral intersects with the x -axis and y -axis
- b)** Find the equation of the line going through the first intersection point with both axes
- c)** Find the angle of intersection for the line from b) with the x -axis and y -axis

Extension

Q7. Is it possible to work out the angle between skew lines and if so what does it represent

Hint: Consider previous questions

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